LC 287 Find the duplicate Number

Approach:

- Bit manipulation by XOR gets rejected hence use Floyd's cycle detection method- Treat the array as a linked list where nums[i] points to the next index. The duplicate creates a cycle.
- Initiate two pointers slow(moves 1 step) and fast(moves 2 step)
- When they meet at an index , initiate fast to first element and move both of them- one step at a time
- Point where they meet is the duplicate number, return it
  - Or
- We can also sort the array and return the element by comparing the neighbour nums[i]==nums[i+1]-brute force works too

TUF SDE Sheet Arrays-Part 2 Q5- Find repeated and missing number

Approach:

- To find the missing number and repeated number,
  - Find the sum of all elements in array and sum of all N(len of arr) natural numbers and find the difference X-Y
  - Find the sum of squares of all array elements and sum of squares of all N natural numbers and find the difference-X^2-Y^2
- Find X+Y from X^2-Y^2/X-Y as X^2-Y^2=(X-Y)(X+Y)
- Now solve X+Y and X-Y that gives X=repeated number and Y=Missing number

TUF SDE Sheet Arrays-Part 2 Q6- Find repeated and missing number

Approach:

- We solve it using **modified merge sort**
- Divide the array into halves recursively
- Count inversion sin left half and right half
- Count cross inversions while merging(when an elemnt from right half is smaller than right half)
- Combine all counts